

ENERGY INFRASTRUCTURE ECONOMICS, POWER, FUELS AND ENERGY SECURITY

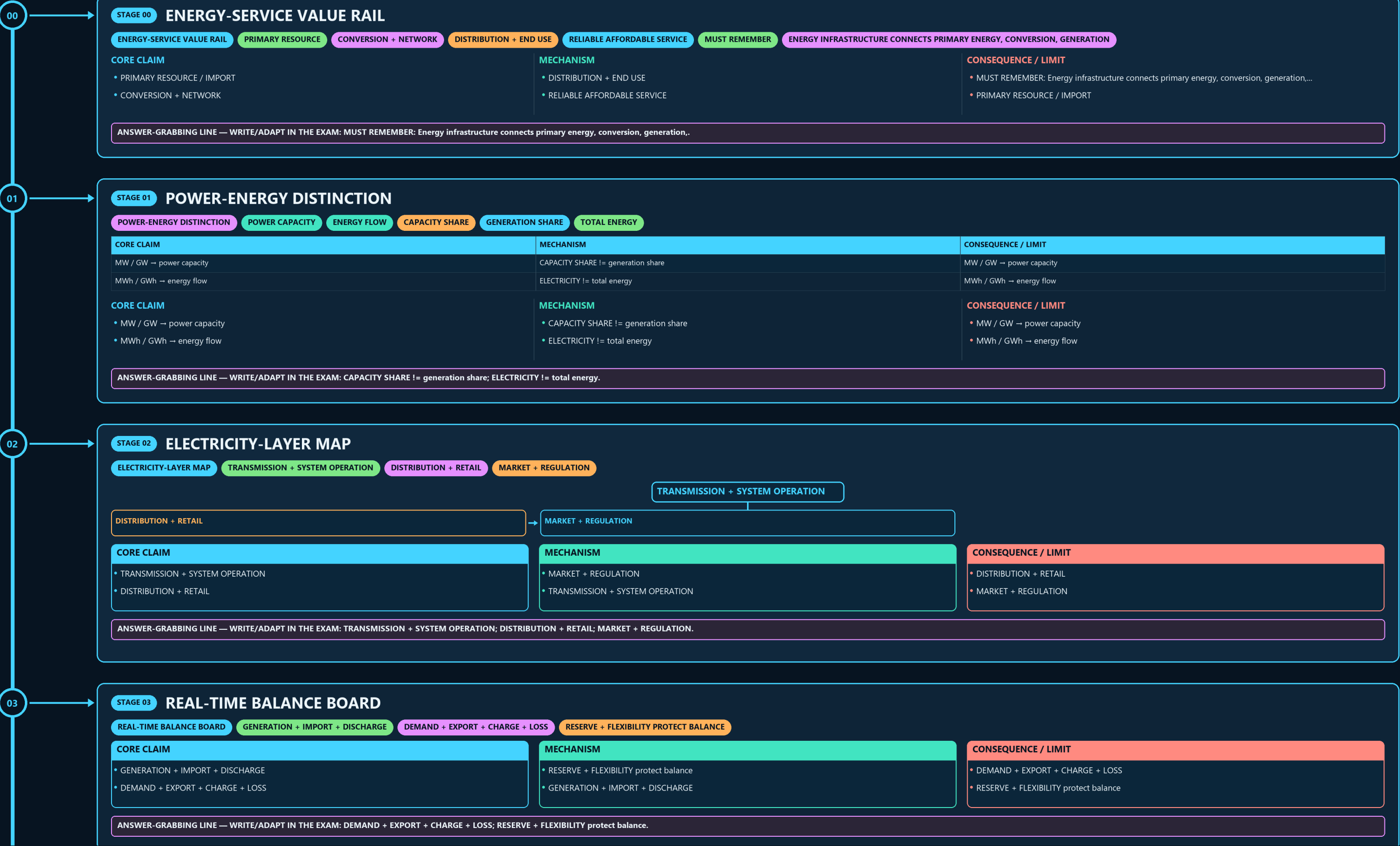
ENERGY-SERVICE VALUE RAIL → POWER-ENERGY DISTINCTION → ELECTRICITY-LAYER MAP → REAL-TIME BALANCE BOARD → INSTITUTIONAL PERIMETER → TARIFF-FINANCE SPLIT

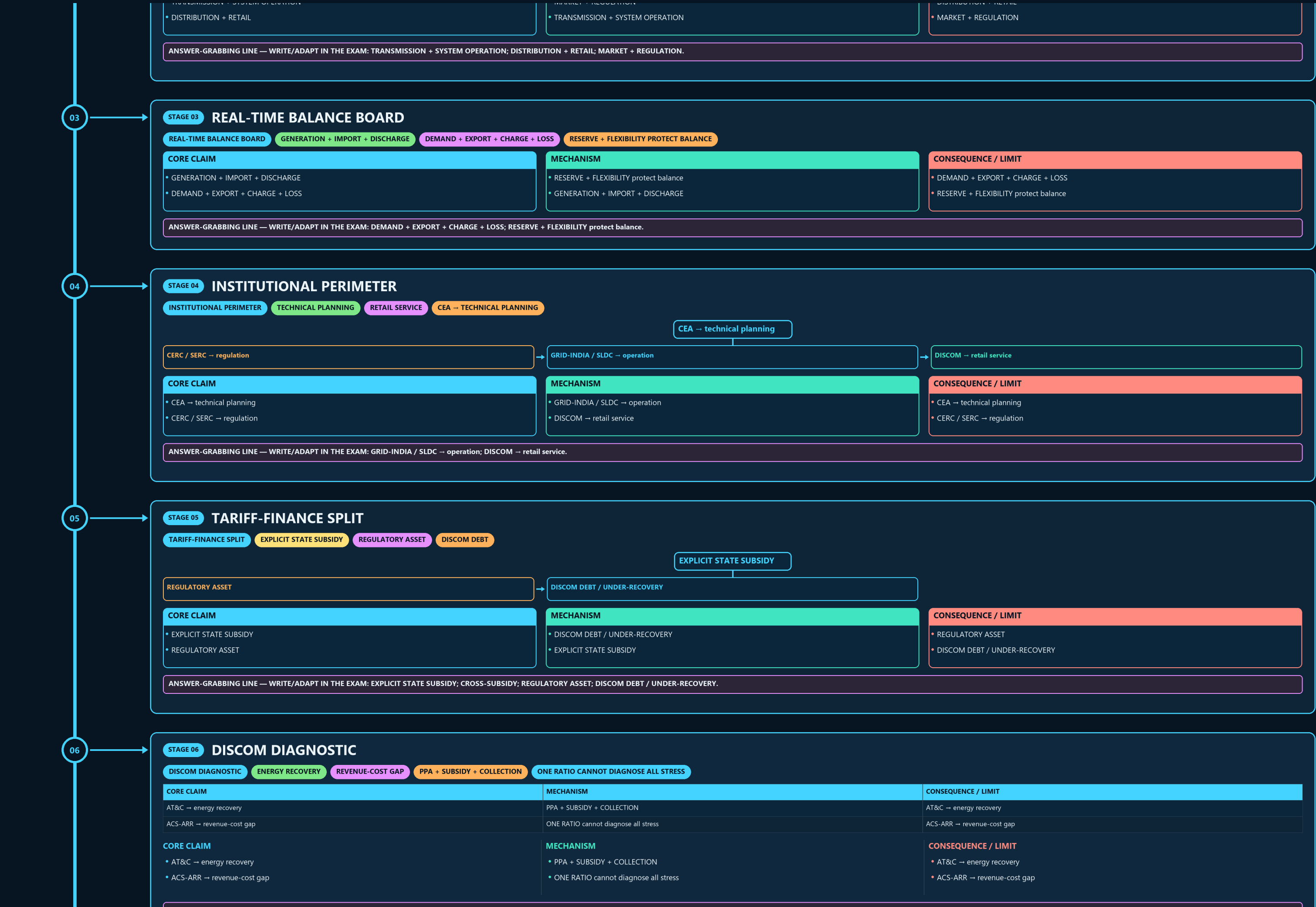
Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles





ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXPLICIT STATE SUBSIDY; CROSS-SUBSIDY; REGULATORY ASSET; DISCOM DEBT / UNDER-RECOVERY.

06

STAGE 06 DISCOM DIAGNOSTIC

DISCOM DIAGNOSTIC ENERGY RECOVERY REVENUE-COST GAP PPA + SUBSIDY + COLLECTION ONE RATIO CANNOT DIAGNOSE ALL STRESS

CORE CLAIM

AT&C → energy recovery
ACS-ARR → revenue-cost gap

MECHANISM

PPA + SUBSIDY + COLLECTION
ONE RATIO cannot diagnose all stress

CONSEQUENCE / LIMIT

AT&C → energy recovery
ACS-ARR → revenue-cost gap

CORE CLAIM

- AT&C → energy recovery
- ACS-ARR → revenue-cost gap

MECHANISM

- PPA + SUBSIDY + COLLECTION
- ONE RATIO cannot diagnose all stress

CONSEQUENCE / LIMIT

- AT&C → energy recovery
- ACS-ARR → revenue-cost gap

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PPA + SUBSIDY + COLLECTION; ONE RATIO cannot diagnose all stress.

07

STAGE 07 PETROLEUM PERIMETER

PETROLEUM PERIMETER PIPELINE + STORAGE REFINING + MARKETING PNGRB EXCLUDES CRUDE + GAS PRODUCTION

CORE CLAIM

- UPSTREAM → production
- MIDSTREAM → pipeline + storage

MECHANISM

- DOWNSTREAM → refining + marketing
- PNGRB excludes crude + gas production

CONSEQUENCE / LIMIT

- UPSTREAM → production
- MIDSTREAM → pipeline + storage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DOWNSTREAM → refining + marketing; PNGRB excludes crude + gas production.

08

STAGE 08 ENERGY-SECURITY DIMENSIONS AND RESILIENCE PENTAGON

ENERGY-SECURITY DIMENSIONS AND RESILIENCE PENTAGON PHYSICAL AVAILABILITY INFRASTRUCTURE ACCESSIBILITY USER AFFORDABILITY SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE

PHYSICAL AVAILABILITY

INFRASTRUCTURE ACCESSIBILITY

USER AFFORDABILITY

SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE

CORE CLAIM

- PHYSICAL AVAILABILITY
- INFRASTRUCTURE ACCESSIBILITY

MECHANISM

- USER AFFORDABILITY
- SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE

CONSEQUENCE / LIMIT

- PHYSICAL AVAILABILITY
- INFRASTRUCTURE ACCESSIBILITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INFRASTRUCTURE ACCESSIBILITY; USER AFFORDABILITY; SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE.

09

STAGE 09 RENEWABLE-INTEGRATION CHAIN

RENEWABLE-INTEGRATION CHAIN VARIABLE CAPACITY GRID + FORECAST FLEXIBILITY + STORAGE RELIABLE GENERATION VALUE CLOSE DISTINCTION CAPACITY IN MW IS NOT GENERATION IN MWH, PLANT

VARIABLE CAPACITY

GRID + FORECAST

FLEXIBILITY + STORAGE

RELIABLE GENERATION VALUE

CORE CLAIM

- VARIABLE CAPACITY
- GRID + FORECAST

MECHANISM

- FLEXIBILITY + STORAGE
- RELIABLE GENERATION VALUE

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Capacity in MW is not generation in MWh, plant load factor is not...
- VARIABLE CAPACITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Capacity in MW is not generation in MWh, plant load factor is not.

09

STAGE 09 RENEWABLE-INTEGRATION CHAIN

RENEWABLE-INTEGRATION CHAIN

VARIABLE CAPACITY

GRID + FORECAST

FLEXIBILITY + STORAGE

RELIABLE GENERATION VALUE

CLOSE DISTINCTION

CAPACITY IN MW IS NOT GENERATION IN MWh, PLANT

VARIABLE CAPACITY

GRID + FORECAST

FLEXIBILITY + STORAGE

RELIABLE GENERATION VALUE

CORE CLAIM

- VARIABLE CAPACITY
- GRID + FORECAST

MECHANISM

- FLEXIBILITY + STORAGE
- RELIABLE GENERATION VALUE

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Capacity in MW is not generation in MWh, plant load factor is not...
- VARIABLE CAPACITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Capacity in MW is not generation in MWh, plant load factor is not.

10

STAGE 10 STATUS AND UNIT LADDER

STATUS AND UNIT LADDER

INSTALLED CAPACITY

GENERATION + SERVICE OUTCOME

EVIDENCE LIMIT

STATE FUEL/TECHNOLOGY, STOCK OR FLOW, UNIT

ANNOUNCED / BILL

APPROVED / SANCTIONED / OUTLAY

INSTALLED CAPACITY

GENERATION + SERVICE OUTCOME

CORE CLAIM

- ANNOUNCED / BILL
- APPROVED / SANCTIONED / OUTLAY

MECHANISM

- INSTALLED CAPACITY
- GENERATION + SERVICE OUTCOME

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State fuel/technology, stock or flow, unit,...
- ANNOUNCED / BILL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State fuel/technology, stock or flow, unit..

11

STAGE 11 ENERGY-ECONOMICS ANSWER SPINE

ENERGY-ECONOMICS ANSWER SPINE

DEFINE CARRIER + UNIT + LAYER

TRACE NETWORK + TARIFF + REGULATOR

TEST SECURITY + TRANSITION

CONCLUDE SERVICE, NOT HEADLINE CAPACITY

DEFINE carrier + unit + layer

TRACE network + tariff + regulator

TEST security + transition

CONCLUDE service, not headline capacity

CORE CLAIM

- DEFINE carrier + unit + layer
- TRACE network + tariff + regulator

MECHANISM

- TEST security + transition
- CONCLUDE service, not headline capacity

CONSEQUENCE / LIMIT

- DEFINE carrier + unit + layer
- TRACE network + tariff + regulator

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TEST security + transition; CONCLUDE service, not headline capacity.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

DISTRIBUTIONAL AND TRANSITION COST

OVERCONSUMPTION IF VARIABLE PRICE IS TOO LOW

EXCLUSION IF FIXED CHARGE IS TOO HIGH

REVENUE INSTABILITY IF FIXED NETWORK COST IS RECOVERED MAINLY

SHIFTS BURDEN TO FUTURE CONSUMERS

CAPACITY NO LONGER NEEDED

OLD HIGH TARIFFS AFTER TECHNOLOGY COST FALLS

OPTIONAL SOURCE DEPTH

- distributional and transition cost
- overconsumption if variable price is too low;
- exclusion if fixed charge is too high;
- revenue instability if fixed network cost is recovered mainly per kWh while rooftop/self

ADVANCED DEBATE

- shifts burden to future consumers;
- capacity no longer needed;
- old high tariffs after technology cost falls;
- fuel/transport assumptions that later fail.

LEGACY / NUANCE

- performance and availability incentives;
- transmission readiness before commissioning.
- suppress wholesale energy prices;
- reduce thermal running hours;

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.